GOPURAM® BRAND

M/S. M. S. P. VELAYUTHA NADAR
SREE MURUGAN TILE WORKS
CLAY FLOORING, ROOFING AND DECORATIVE TILES
10/712, Surandai Road, Pavoorchatram - 627808. Tenkasi District, Tamil Nadu, India.

GSTIN: 33AAGFM1766R1ZR
PAN: AAGFM1766R
IEC: 3509007026
+91-4633-250924
www.TileWorks.in
kla@tileworks.in

TO WHOM IT MAY CONCERN

This is to certify that **K. L. Mithunvel (23BMH1029)**, **Kanigalpula Teja Pavan Sai Adinarayana Gupta** (23BMH1007), and **Madhumitha Vemula** (23BMH1019), students of Vellore Institute of Technology, Chennai, have undertaken the project titled "Automated Tile Inspection, Grading and Sorting System", under the guidance of Dr. Jegadeeshwaran R (50204), in relation to the inspection and handling requirements of Sree Murugan Tile Works (SMTW).

The project team has approached the work with dedication and sincerity. The work presented to us includes the following contributions and outcomes:

1. Industrial process study and problem definition. Examined the manual inspection, tap-testing, grading, sorting and packing workflow, and identified opportunities to improve consistency, traceability and handling efficiency.

2. System design and documentation. Prepared the project charter, system architecture, requirements, subsystem designs, electrical and communication plans, and a phased implementation strategy for an integrated tile-inspection and sorting line.

3. Camera-based inspection prototype. Developed computer-vision software to detect and count tiles, examine visible cracks and chipped corners, and produce grades and inspection records. The camera pipeline was evaluated on five conveyor recordings containing 92 tile crossings, with 88 tiles carrying usable human reference labels.

4. Image-based classification and evaluation. Explored and benchmarked tile-grading models, including a reported 82.9% result on a 76-image test set for a particular classification experiment. This test result is not a verified factory-wide inspection accuracy.

5. Acoustic inspection development. Designed a controlled-impact, dual-solenoid gravity-drop tap mechanism and developed software for audio capture, feature extraction, tap sequencing and acoustic grading. Hardware integration and validation with production tiles remain further development activities.

6. Robotic pick-and-place development. Designed the three-axis Cartesian gantry control approach, including motion commands, end-effector control, homing and safety logic, and developed simulation, firmware and a 3D digital twin. Full automated sorting on an assembled production line has not yet been demonstrated.

For Sree Murugan Tile Works

20/10/2026

Partner

GOPURAM® BRAND

M/S. M. S. P. VELAYUTHA NADAR
SREE MURUGAN TILE WORKS
CLAY FLOORING, ROOFING AND DECORATIVE TILES
10/712, Surandai Road, Pavoorchatram - 627808. Tenkasi District, Tamil Nadu, India.

GSTIN: 33AAGFM1766R1ZR
PAN: AAGFM1766R
IEC: 3509007026
+91-4633-250924
www.TileWorks.in
kla@tileworks.in

7. Station coordination and tile traceability. Developed a modular processing and communication approach to associate camera and acoustic observations with individual tiles, combine inspection results into a final decision, and support acknowledgement, exception handling and production records.

8. Monitoring and operational software. Developed station and system monitoring interfaces for displaying inspection status, tile counts, grading outcomes, faults and diagnostic information, with software demonstrations and tests completed.

9. Electrical, sensing and safety engineering. Prepared controller and sensor connections, motor-driver and power arrangements, communication plans, and fail-safe provisions including emergency stops, watchdogs and detection of missing or faulty measurements.

10. Testing, feasibility and implementation planning. Performed software tests and simulation-based checks; assessed hardware choices, preliminary costs and staged deployment requirements, including vacuum handling considerations for clay tiles of varying shapes, masses and surface conditions.

These outcomes represent design work and partially validated prototypes intended to support future industrial automation. They should not be interpreted as certification of a fully commissioned production system at SMTW.

We appreciate the team's efforts and wish them success in their future endeavours.

Authorised Signatory

For Sree Murugan Tile Works

Partner

Name and Designation: K LAKSHMI ANAND - PARTNER
Date: 20/10/2026